

Experiment No-7

Even and odd parts of signal and sequence & Real and imaginary parts of Signal

AIM: Finding even and odd part of the signal and sequence and also find real and imaginary parts of signal.

Software Required: Matlab software

Theory: One of characteristics of signal is symmetry that may be useful for signal analysis. Even signals are symmetric around vertical axis, and Odd signals are symmetric about origin.

Even Signal: A signal is referred to as an even if it is identical to its time-reversed counterparts;

$$x(t) = x(-t).$$

Odd Signal: A signal is odd if $x(t) = -x(-t)$.

An odd signal must be 0 at $t=0$, in other words, odd signal passes the origin.

Using the definition of even and odd signal, any signal may be decomposed into a sum of its even part, $x_e(t)$, and its odd part, $x_o(t)$, as follows

Even and odd part of a signal: Any signal $x(t)$ can be expressed as sum of even and odd components i.e.,

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{1}{2} \{x(t) + x(-t)\},$$

$$x_o(t) = \frac{1}{2} \{x(t) - x(-t)\}.$$

$$x(t) = x_e(t) + x_o(t)$$

$$= \frac{1}{2} \{x(t) + x(-t)\} + \frac{1}{2} \{x(t) - x(-t)\}$$

Program:

```
clc;
close all;
clear all;

%Even and odd parts of a signal
t=0:.001:4*pi;
x=sin(t)+cos(t);           % x(t)=sint(t)+cos(t)
subplot(2,2,1)
plot(t,x)
xlabel('t');
ylabel('amplitude');
title('input signal');
```

```

y=sin(-t)+cos(-t); % y(t)=x(-t)
subplot(2,2,2)
plot(t,y)
xlabel('t');
ylabel('amplitude');
title('input signal with t= -t');
even=(x+y)/2;
subplot(2,2,3)
plot(t,even)
xlabel('t');

```

```

ylabel('amplitude')
title('even part of the signal')
odd=(x-y)/2;
subplot(2,2,4)
plot(t,odd)
xlabel('t');
ylabel('amplitude');
title('odd part of the signal');

```

% Even and odd parts of a sequence

```

x1=[0,2,-3,5,-2,-1,6];
n=-3:3;
y1= fliplr(x1);%y1(n)=x1(-n)
figure;
subplot(2,2,1);
stem(n,x1);
xlabel('n');
ylabel('amplitude');
title('input sequence');
subplot(2,2,2);
stem(n,y1);
xlabel('n');
ylabel('amplitude');
title('input sequence with n= -n');
even1=.5*(x1+y1);
odd1=.5*(x1-y1);

```

% plotting even and odd parts of the sequence

```

subplot(2,2,3);
stem(n,even1);
xlabel('n');

```

```

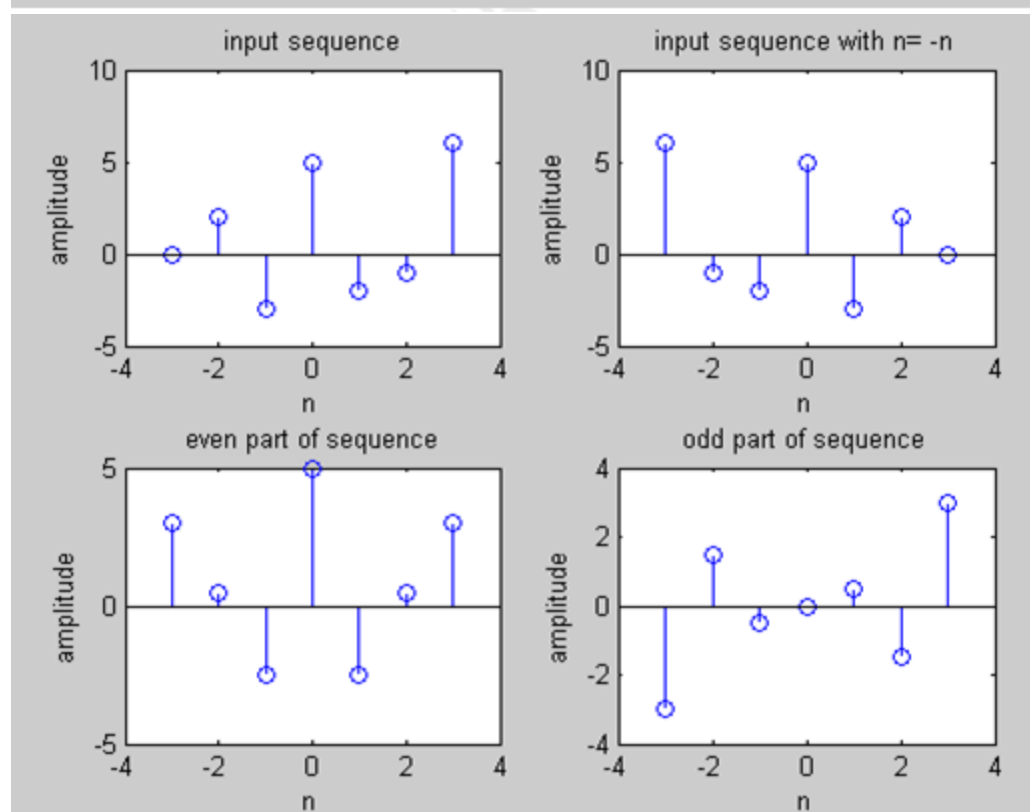
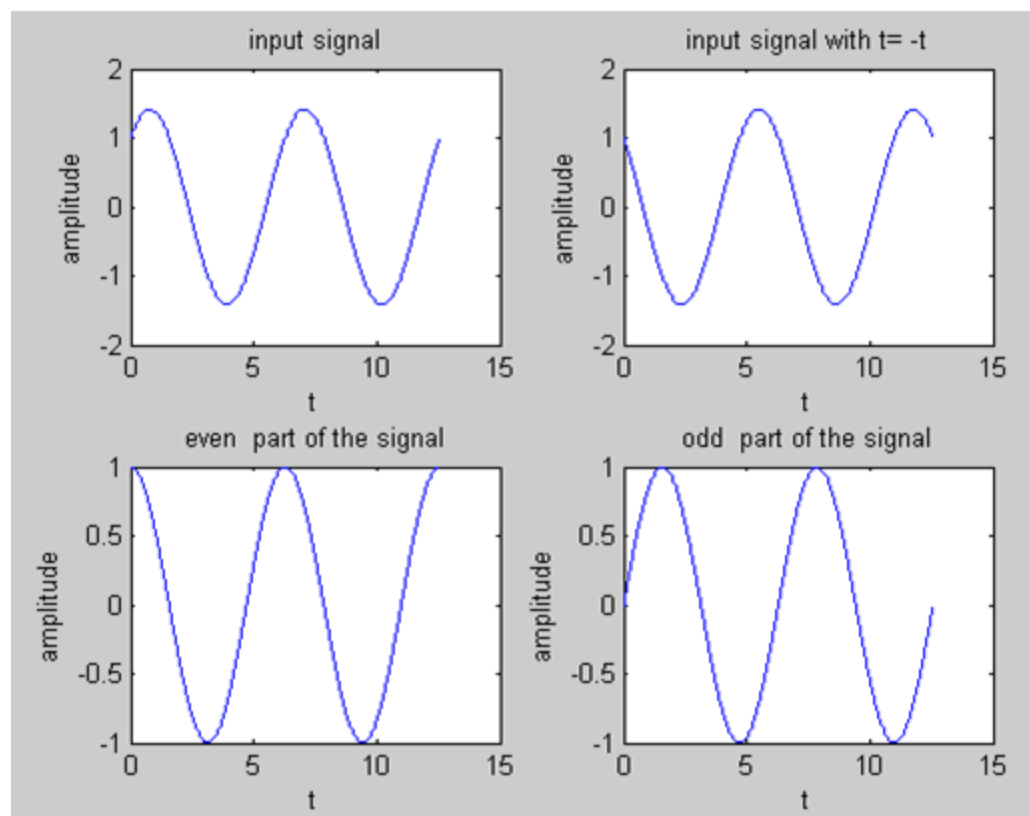
ylabel('amplitude');
title('even part of sequence');
subplot(2,2,4);
stem(n,odd1);
xlabel('n');
ylabel('amplitude');
title('odd part of sequence');

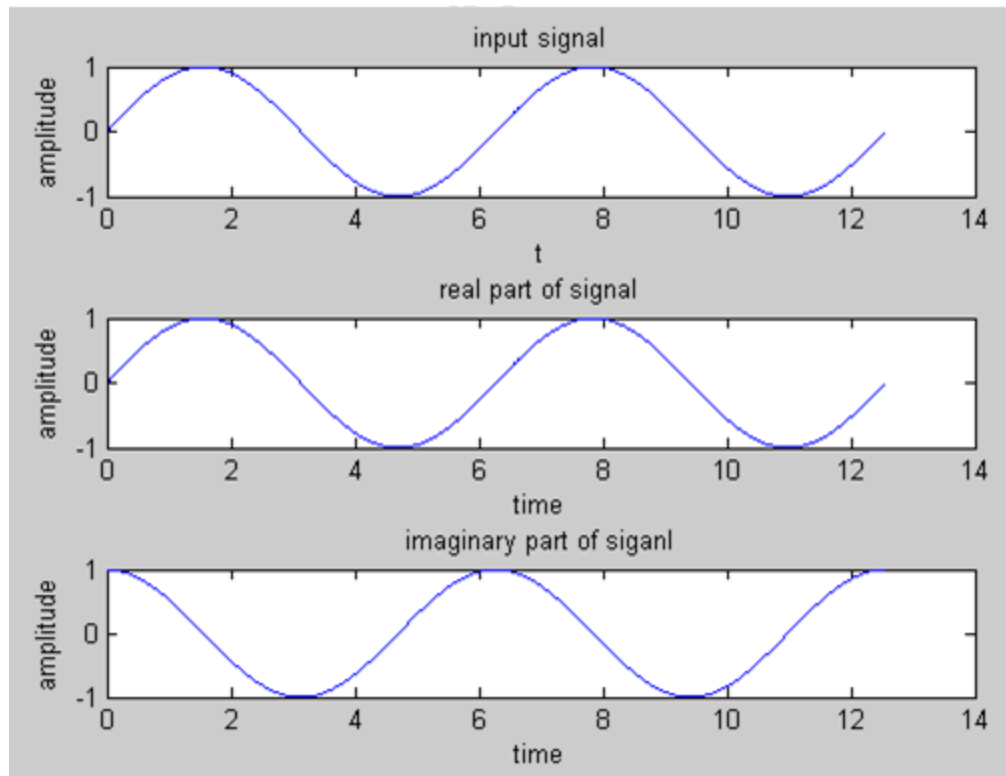
% plotting real and imaginary parts of the signal
x2=sin(t)+j*cos(t);
figure;
subplot(3,1,1);
plot(t,x2);
xlabel('t');
ylabel('amplitude');
title('input signal');
subplot(3,1,2);
plot(t,real(x2));
xlabel('time');
ylabel('amplitude');
title('real part of signal');
subplot(3,1,3);
plot(t,imag(x2));
xlabel('time');
ylabel('amplitude');
title('imaginary part of signal');

```

RESULT: Even and odd part of the signal and sequence, real and imaginary parts of signal are computed.

Output:





VIVA QUESTIONS:-

1. What is the formula to find odd part of signal?
2. What is Even Signal?
3. What is Odd Signal?
4. What is the formula to find even part of signal?
5. What is the difference b/w stem & plot?